Two-Regime Models for Daily Soil-Moisture Estimation in Washington

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Summer research showcase · 2026

Background and research question

Why soil moisture estimation?

Soil moisture is the amount of water held in soil. Daily estimates help with irrigation planning and drought monitoring. They show how moisture changes after rain and during dry periods, including places with few direct measurements.

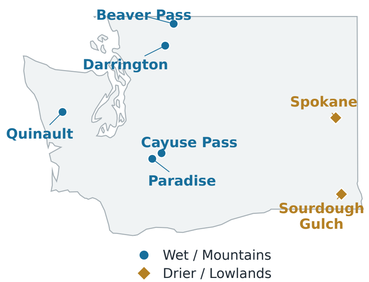
Ground sensors measure moisture at individual sites. Satellite observations cover larger areas but are intermittent, with some retrievals missing. We combine satellite, weather, and site data for daily estimates. Earlier work produced the pipeline and a global XGBoost model.

Summer research question

Can two specialists improve estimates across Washington's contrasting conditions?

Our comparison uses the same evaluation conditions for both models. For known stations, we fix a single specialist; tests at new sites examine how missing satellite inputs affect assignment.

Study area and data



Regional contrast

Our stations span the coast, Cascade mountains, and drier eastern Washington. They let us study how soil moisture relates to inputs under different conditions. Station holdouts test whether the models can estimate moisture at a new site.

Figure 1. The seven training stations and their fitted groups. Colors identify station assignments, with no geographic boundaries implied.

NOAA USCRN and SNOTEL stations provide measured soil moisture. Satellite inputs include SMAP soil moisture, Sentinel-1 radar backscatter from the land surface, and Sentinel-2 surface reflectance and indices of vegetation greenness and canopy water content. We also use MODIS land-surface temperature and vegetation indices, ERA5-Land weather, and terrain, climate, land-cover, and soil descriptors.

The pipeline puts observations on a daily grid, fills gaps, smooths the time series, and calculates features describing recent satellite readings and weather.

2017 to 2022
Final fitting period → **2023 to 2025**
Later-year test

Development uses 2017 to 2020 for training and 2021 to 2022 for validation. The final fit combines both periods. We test later years and refit without each station in turn to evaluate new Washington locations.

Two-regime modeling

A global model learns one input-to-soil-moisture relationship from all stations. We divide observations into two groups and fit an XGBoost model to each. Each specialist predicts daily volumetric soil moisture at 5 cm depth (m³/m³).

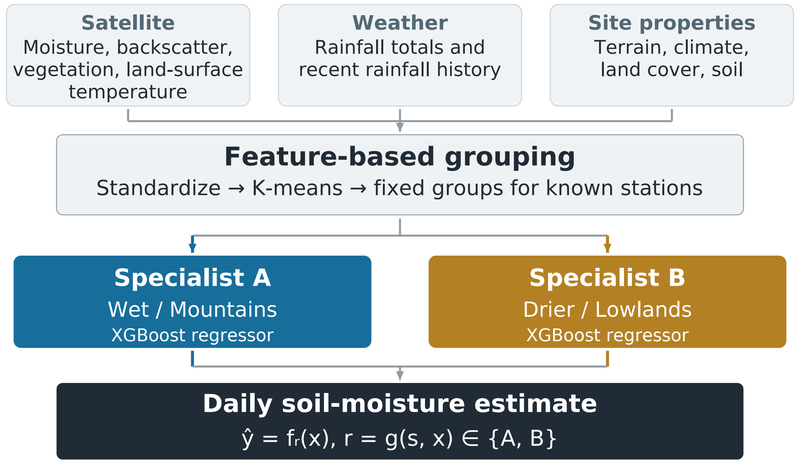


Figure 2. Each observation uses one specialist. x denotes the inputs, s the station, g the assignment rule, and f_r the selected model.

1. We fill missing router inputs with their fitting-data means and standardize features to comparable scales. K-means finds two cluster centers; a new observation goes to the nearest center. The router uses inputs alone, without measured target labels.

2. For a station included in fitting, its most common group selects one specialist for all its observations. New stations receive an assignment each day. Missing routing inputs can trigger the precipitation fallback.

3. Both approaches use identical features, XGBoost settings, evaluation observations, and random seeds. Each specialist trains on its assigned group. Seeds control training randomness; repeated fits measure training variation, leaving uncertainty across new sites and seasons unmeasured.

Regime interpretation

Wet / Mountains

5 training stations

Western and Cascade sites: more rain and higher SMAP soil moisture.

Wettest-month rain: 391 mm
SMAP soil moisture: 0.414 m³/m³

Drier / Lowlands

2 training stations

Spokane and Sourdough Gulch: less rain and lower SMAP soil moisture.

Wettest-month rain: 70 mm
SMAP soil moisture: 0.192 m³/m³

The groups differ mainly in rainfall and SMAP soil moisture. The wetter group includes lowland and mountain stations, so elevation alone does not distinguish them. Values above are input-data medians. Two groups worked for these stations; other deployments should choose their group count using local data.

Evaluation and results

+0.0320
Later-year R^2 gain

+0.0581
Station-holdout R^2 gain

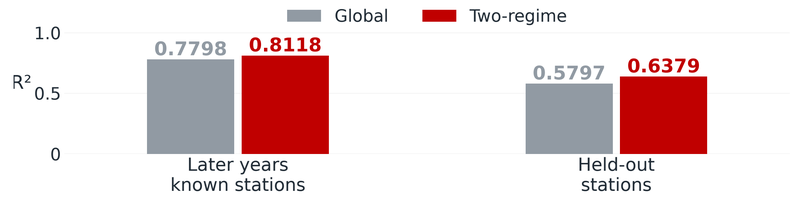


Figure 3. Mean R^2 from 30 later-year runs and five per held-out station, with matched seeds. Higher is better.

Two-regime R^2 is higher in all 30 later-year runs. Leaving out stations also favors specialization on average, although improvements vary by location.

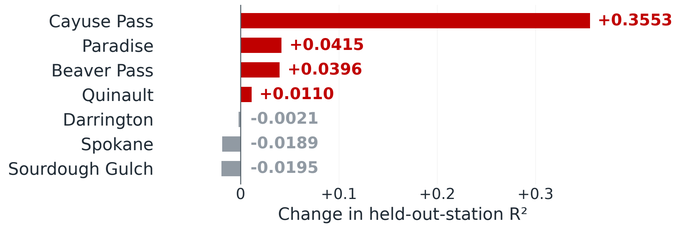


Figure 4. Mean R^2 change: four gains and three losses (five runs).

Cayuse Pass has the largest R^2 increase. Leaving out either eastern station reduces the smaller group's training coverage.

ECE sensor-site evaluation

Missing SMAP inputs at five new sites deployed by Seattle University's ECE team triggered automatic assignment using an index of recent rainfall. We used Washington validation data to choose a specialist for each rainfall class.

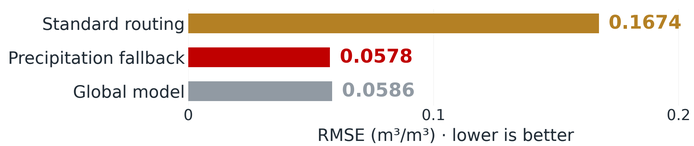


Figure 5. Pooled ECE RMSE. Rainfall fallback and global model errors are similar; lower is better.

For all 150 daily observations from July 20 to August 19, 2026, the fallback selected the same specialist. Full-season performance at these ECE sites remains untested.

Discussion and next steps

These gains depend on reliable assignment and training sites that cover the conditions encountered. We will test more Washington sites and seasons with ECE sensors, including changes in satellite availability.